Confidence Intervals with the T-procedure

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Bessel's correction; Confidence intervals.
- **Leads to:** T-tests: one-sample, two-sample, paired; Correlation significance test.
- **Compare with:** Normal distribution.

1. Overview

Key point: When the population standard deviation σ is unknown, we use the sample standard deviation s and take the critical value from **Student's t-distribution** (G-1906) instead of the standard normal curve.

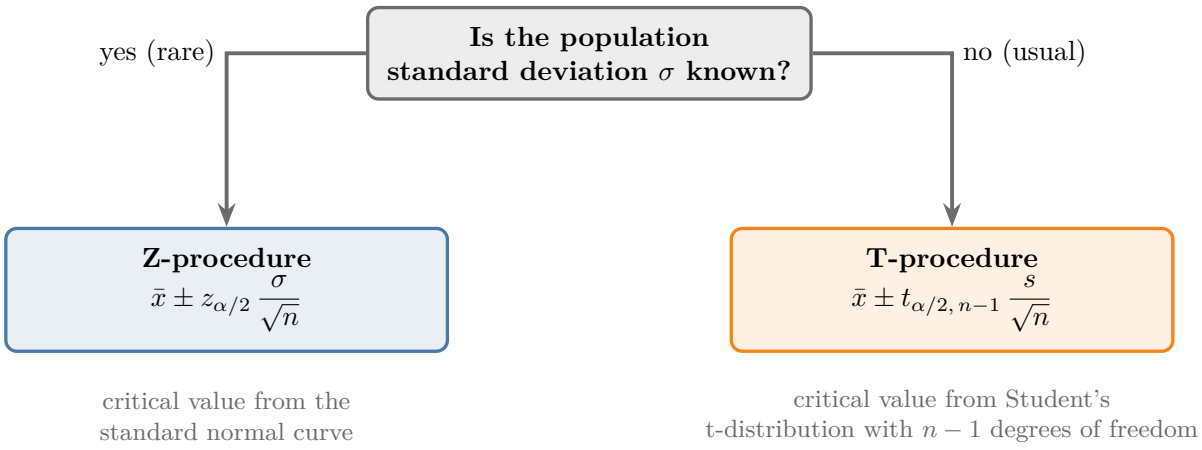


Figure 1: Choosing between the z-procedure and the t-procedure

Earlier we built the **confidence interval** (a range that is likely to contain the population mean) $\bar{x} \pm z_{\alpha/2}\, \sigma/\sqrt{n}$, where $z_{\alpha/2}$ is the z critical value (1.96 for 95% confidence); see the z-procedure formula. The z-interval needs σ, the standard deviation of the whole population, which we almost never have. Figure 1 shows the way out: the **t-procedure** (G-1938), the method used in real work.

This Note covers:

- why replacing σ by s changes the distribution (section 3);
- a simulation showing that "z with s" fails and "t with s" works (section 4);
- Student's t-distribution and its **degrees of freedom** (G-578) (section 5);
- the t-interval and t critical values (section 6);
- the assumptions of the t-procedure (section 7);
- a t-interval for the mean Titanic fare (section 8).

2. Why σ is rarely known

> **Key point:** If we do not know the population mean, we almost never know the population standard deviation either.

The z-procedure has a hidden weakness. We build a confidence interval because we do not know the mean age μ of a channel's 77,000 subscribers. But σ is computed from the same 77,000 ages: if we cannot get μ, we cannot get σ either.

So in practice the z-procedure is rarely usable. The **t-procedure** (G-1938) needs no population parameter at all: everything comes from the sample.

3. Replacing σ by s

> **Key point:** With s in place of σ, the standardized sample mean no longer follows the standard normal distribution but **Student's t-distribution** (G-1906).

The natural fix is to use the next best thing. We use $\bar{x}$ when μ is unknown; so we use the sample standard deviation s (dividing by $n-1$, see the sample variance) when σ is unknown.

The z-procedure rested on the standardized sample mean $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$, which follows the standard normal distribution. Putting s in place of σ gives a new quantity.

1. **In words:** standardize the sample mean with the sample's own standard deviation.
2. **Formula:**
$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$
3. **Example:** a sample of $n = 10$ ages with $\bar{x} = 28$ and $s = 15$, if the true mean were $\mu = 25$:
$$T = \frac{28 - 25}{15/\sqrt{10}}$$
$$T = \frac{3}{4.743} = 0.63$$

T looks like Z, but it is not standard normal. In Z only the numerator varies from sample to sample. In T the denominator varies too: a sample with an unusually small s gives a large T. This extra uncertainty makes extreme values of T more common than extreme values of Z.

4. Why "z with s" fails

> **Key point:** Using s with the z critical value gives intervals that are too narrow: for $n = 10$, a "95%" interval contains μ only 91.9% of the time; the t critical value restores 95%.

The tempting fix is to keep the z-interval and simply write s where σ was. A simulation shows what that costs. As in seeing it by simulation, we pretend to know the population, a normal one with $\mu = 50$ and $\sigma = 15$, and draw samples of $n = 10$. From each sample we build three "95%" intervals:

- **z with σ:** $\bar{x} \pm 1.96\,\sigma/\sqrt{n}$, the z-procedure, possible only because we invented the population;
- **z with s:** $\bar{x} \pm 1.96\,s/\sqrt{n}$, the tempting shortcut;
- **t with s:** $\bar{x} \pm 2.262\,s/\sqrt{n}$, the **t-procedure** (G-1938) of section 6, which uses a larger critical value.

In Figure 2 the intervals arrive in batches. Watch the orange intervals, the misses, and the count in each title: the middle panel collects misses fastest.

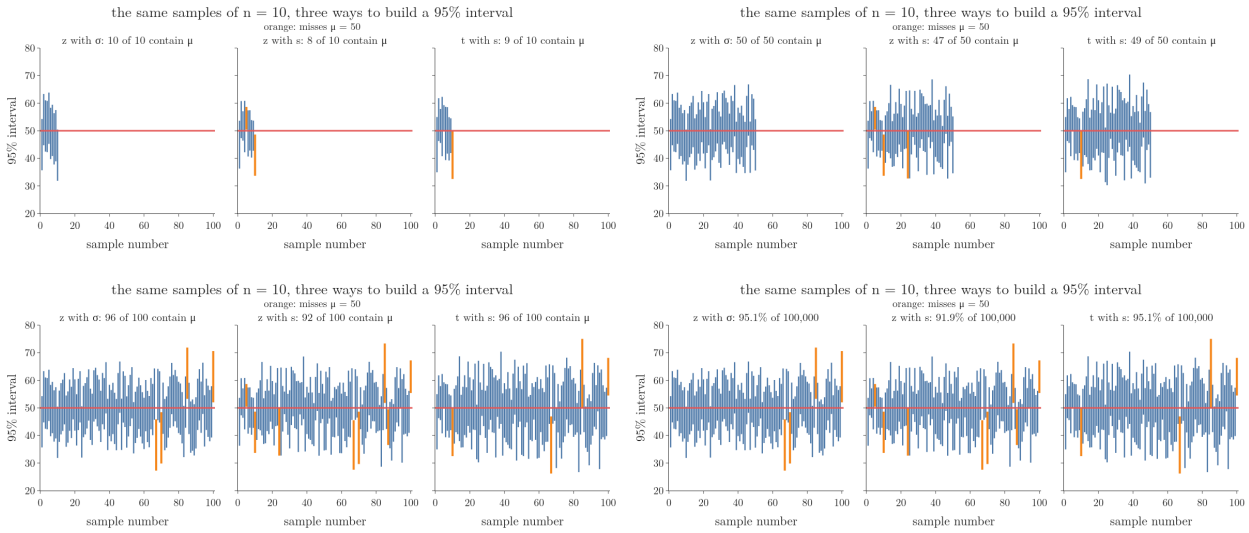


Figure 2: The same samples of n = 10 with three 95% intervals each: z with the true sigma, z with s, and t with s; misses in orange, running counts in the titles, coverage over 100,000 samples in the last frame (idea after Khan Academy, "Simulation showing value of t statistic")

Over 100,000 samples, the "95%" z-with s intervals contain μ only 91.9% of the time, while z with the true σ and t with s both reach 95.1% (Figure 2, last frame). Repeating more often does not help: the z intervals stay around 92%, because they are built from the wrong distribution. They are systematically too short and skip the true mean more often than promised.

Sample size n	z with s	t with s
5	87.7%	95.0%
10	91.9%	95.1%
30	94.2%	95.2%
100	94.7%	95.0%

The gap is large for small samples and fades as n grows, exactly as the t-curve approaches the normal curve. The rule is simple:

- σ known: z-procedure, $\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$;
- σ unknown: t-procedure, $\bar{x} \pm t_{\alpha/2,\,n-1}\,s/\sqrt{n}$.

5. Student's t-distribution

> **Key point:** A bell-shaped, symmetric distribution like the standard normal but with fatter tails; its only parameter is the **degrees of freedom** (G-578), $n-1$.

The distribution of T is **Student's t-distribution** (G-1906). The t-distribution was published in 1908 by William Sealy Gosset, a chemist at the Guinness brewery in Dublin who had to draw conclusions from small samples, for example when choosing the best varieties of barley (Student 1908; Zabell 2008). Guinness did not let staff publish under their own names, so Gosset signed the paper "Student", and the name stuck (Zabell 2008). The letter t has no special meaning.

Unlike the normal and log-normal distributions, which describe raw data such as heights or incomes, the t-distribution was derived for this inference problem. The t-distribution describes a statistic, T, not measurements.

5.1 Degrees of freedom

> **Key point:** The t-distribution has one parameter, the degrees of freedom: $df = n-1$ for a sample of size n.

The normal distribution has two parameters, μ and σ. The t-distribution has one, the **degrees of freedom** (G-578) (df):

$$df = n - 1$$

A sample of 50 gives 49 degrees of freedom; a sample of 30 gives 29.

The $n-1$ is the same $n-1$ as in the sample variance. The n deviations $x_i - \bar{x}$ always add up to zero, so once $n-1$ of them are known, the last one is fixed: only $n-1$ of them are free to vary. Bessel's correction (dividing by $n-1$) divides by these $n-1$ free deviations. For a normal population, $(n-1)s^2/\sigma^2$ follows a chi-square distribution (a distribution of sums of squares) with $n-1$ degrees of freedom, and this is why T has $n-1$ degrees of freedom (Casella and Berger 2002, §5.3).

5.2 Shape: fatter tails

> **Key point:** Few **degrees of freedom** (G-578) give fat tails; as df grows, the t-distribution approaches the standard normal, and equals it in the limit.

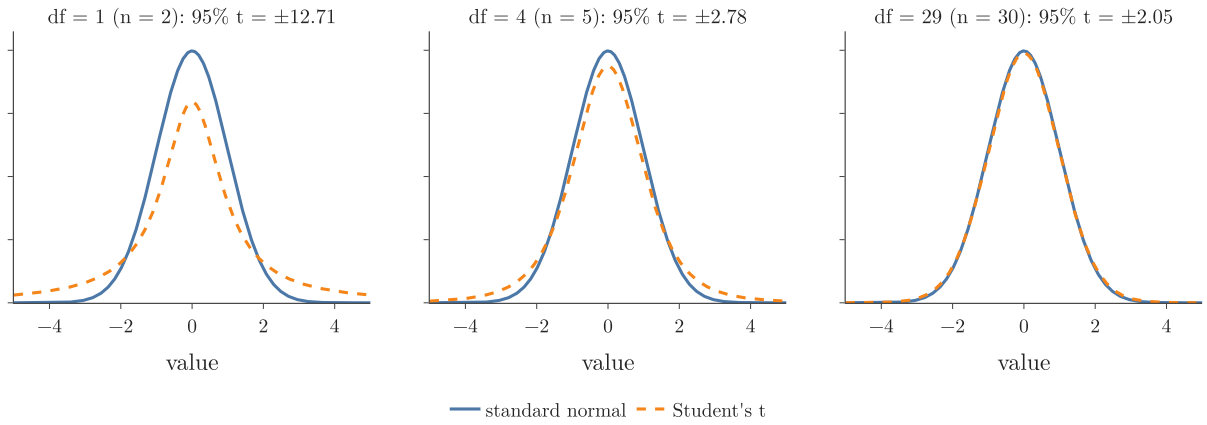


Figure 3: Student's t (dashed) against the standard normal (solid) for 1, 4 and 29 degrees of freedom, with the 95% t critical values (z = ±1.96)

Figure 3 compares the two curves:

- both are bell-shaped, symmetric and centred at 0;
- the t-curve is lower in the middle and higher in the tails: some probability has moved from near the mean to far away;
- with 1 degree of freedom (a sample of 2) the difference is large; with 29 (a sample of 30) the curves almost coincide.

The fat tails are the extra uncertainty of s, put into the shape. The probability of a value above 3 shows it:

Distribution	$P(\text{value} > 3)$
t, $df = 1$	0.1024
t, $df = 4$	0.0200
t, $df = 29$	0.0027
t, $df = 1000$	0.0014
standard normal	0.0013

As n grows, s becomes a reliable estimate of σ, the extra uncertainty disappears, and the t-distribution approaches the standard normal. The t-distribution becomes exactly normal only with infinitely many degrees of freedom, but beyond about 30 to 50 the two are very close.

Figure 4 builds the t-distribution from scratch. For each sample size, we draw 20,000 samples from the normal population of section 4 ($\mu = 50$, $\sigma = 15$) and compute T for each. Watch the red tails beyond ±1.96: they hold 30% of the samples at $n = 2$ and shrink towards 5% as n grows, while the histogram follows the t-curve, not the normal curve.

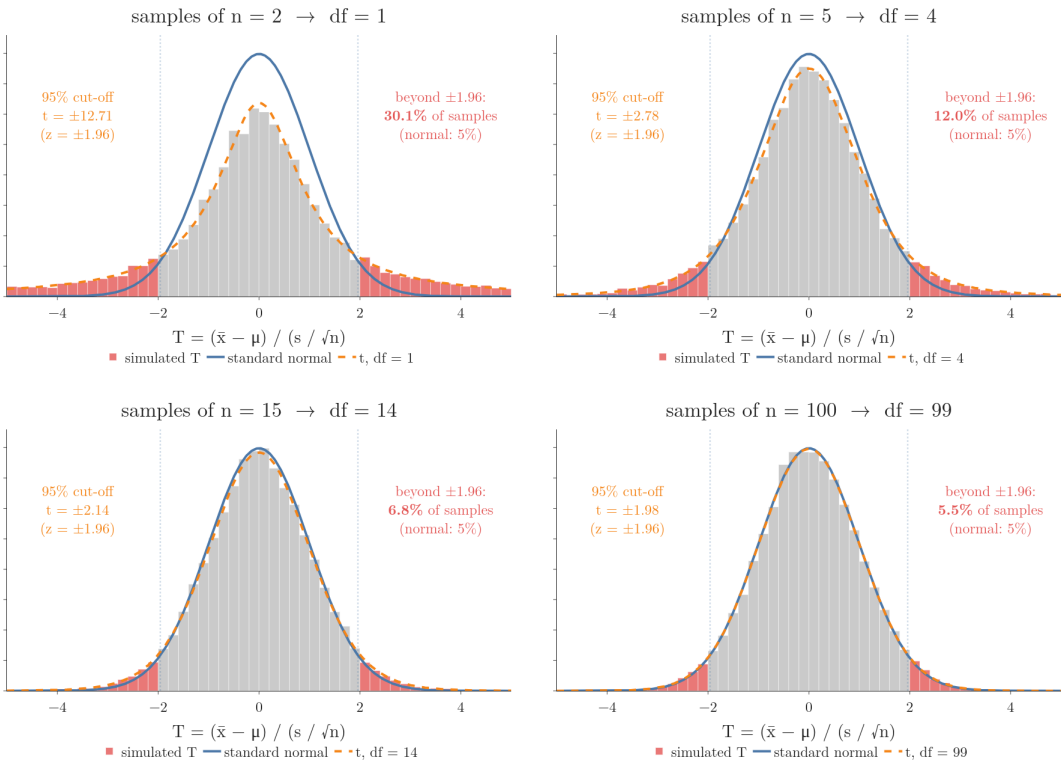


Figure 4: Simulated T values for samples of n = 2 to 100 (histogram) against the standard normal (blue) and the t-curve with df = n - 1 (orange); the share beyond ±1.96 falls from 30% towards 5%

6. The t-procedure formula

> **Key point:** $\bar{x} \pm t_{\alpha/2,\,n-1}\, s/\sqrt{n}$: the z-interval with s for σ and a t critical value for $z_{\alpha/2}$.

Since T follows the t-distribution, the derivation of the z-interval (see where the formula comes from) goes through unchanged with t in place of z.

1. **In words:** go from the sample mean $t_{\alpha/2}$ estimated standard errors down and up, where the standard error (the typical distance of a sample mean from the population mean) is estimated by $s/\sqrt{n}$ and $t_{\alpha/2}$ comes from the t-distribution with $n-1$ **degrees of freedom** (G-578).
2. **Formula:**
$$\bar{x} \pm t_{\alpha/2,\,n-1}\,\frac{s}{\sqrt{n}}$$
3. **Example:** 10 subscribers with $\bar{x} = 28$ years and $s = 15$ years, 95% confidence, $df = 9$, so $t_{0.025,\,9} = 2.262$:
$$\frac{s}{\sqrt{n}} = \frac{15}{\sqrt{10}} = 4.743$$
$$E = 2.262 \times 4.743 = 10.73$$

$$28 - 10.73 = 17.27$$

$$28 + 10.73 = 38.73$$

The interval is 17.27 to 38.73 years. With $z = 1.96$ the margin would be narrower, and, as section 4 shows, too narrow:

$$1.96 \times 4.743 = 9.30$$

$$18.70 \text{ to } 37.30$$

The t critical value is always larger than $z_{\alpha/2}$ for the same confidence level. The fat tails push the cut-offs that leave 2.5% in each tail further out (Figure 3). A wider interval is the honest answer: we are less sure about the spread, so we claim less precision.

6.1 Reading a t-table

> **Key point:** Find the row for $df = n - 1$ and the column for an upper-tail area of $\alpha/2$; for 95%, that is the 0.025 column.

A t-table has one row per **degrees of freedom** (G-578) and one column per upper-tail area. For a two-sided 95% interval, each tail holds 0.025, so we read the 0.025 column (some tables label it "95%, two-sided"). For a sample of 30, $df = 29$ gives 2.045.

df	$t_{0.025}$ (95%, two-sided)	$t_{0.05}$ (one tail of 5%)
1	12.706	6.314
4	2.776	2.132
9	2.262	1.833
29	2.045	1.699
30	2.042	1.697
40	2.021	1.684
60	2.000	1.671
100	1.984	1.660
1000	1.962	1.646
z (normal)	1.960	1.645

Figure 5 draws both columns of the table against the degrees of freedom. Watch the blue curve: it starts far above 1.96 for small samples and sinks towards it, but never below.

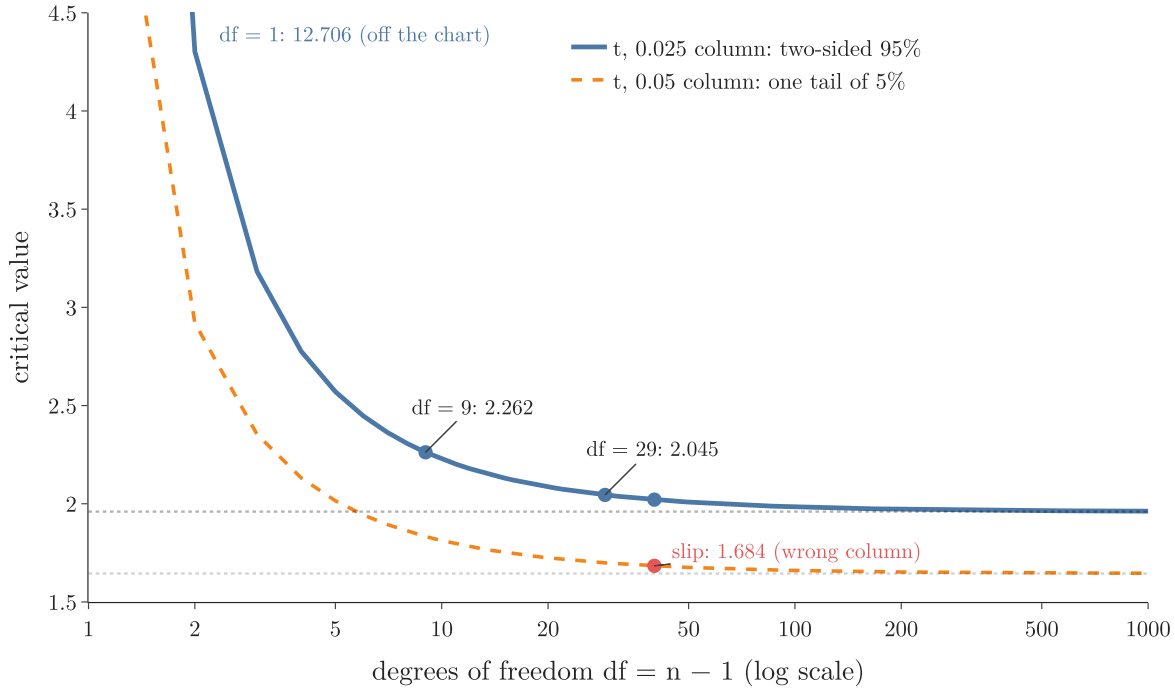


Figure 5: T critical values against degrees of freedom: the 0.025 column (two-sided 95%, blue) and the 0.05 column (one tail of 5%, orange), with z = 1.96 and 1.645 dotted

A common slip is to read the 0.05 column for a 95% interval. For $df = 40$ it gives 1.684 instead of 2.021: that value leaves 5% in **one** tail, which suits a one-sided interval ("the mean is at least ...") but not a two-sided one. The correct value is always above 1.96.

Python: T critical values and t-intervals with scipy.

```
from scipy import stats

stats.t.ppf(0.975, df=29)        # 2.045: two-sided 95%, n = 30
stats.t.ppf(0.95, df=40)         # 1.684: one-sided 95%
# the whole interval in one call
stats.t.interval(0.95, df=9, loc=28, scale=15 / 10 ** 0.5)
# (17.27, 38.73)
```

7. Assumptions of the t-procedure

Key point: A random sample, a roughly normal population or a large sample, and independent observations.

1. **Random sample.** As for the z-procedure: drawn at random and representative.
2. **Normal population, or a large sample.** The population should be approximately normal. If it is not, a sample of more than about 30 lets the central limit theorem make $\bar{x}$ approximately normal anyway (see the central limit theorem).

3. **Independent observations.** An **observation** (G-1374) is one record, here one value in the sample. One observation must not influence another. The ages of randomly chosen subscribers are independent. Feedback ratings may not be: if one person's rating sways the next, the observations are linked, and the **t-procedure** (G-1938) still gives an interval, but one that is too narrow.

> **Extra:** The notebook measures the damage. For samples of 30 from a normal population where each value is tied to the one before (correlation 0.5), the "95%" t-interval contains μ only 74.1% of the time, against 95.2% for independent values.

Figure 6 shows the damage on 60 samples of each kind. Watch the red intervals on the right: with linked values the sample means scatter further from μ than $s/\sqrt{n}$ allows for, so about one interval in four misses.



Figure 6: 60 t-intervals from independent samples (left) and from samples where each value is tied to the one before (right); red intervals miss the mean 50. Coverage over 20,000 samples in the titles

The t-procedure has two strengths:

- it works when σ is unknown, the usual case, because it uses s and the t critical value;
- by the second assumption, for a population that is close to normal, it also works with **very small samples**, such as 5 or 10 values, where the CLT cannot help.

8. Case study: the mean Titanic fare

> **Key point:** One random sample of 30 passengers gives the 95% t-interval 18.57 to 55.72 pounds, which contains the true mean of 33.30 pounds.

We return to the population of Titanic fares: the 1308 known fares of all Titanic passengers, with true mean 33.30 pounds. We know neither μ nor σ (we pretend), so the **t-procedure** (G-1938) is the right tool.

1. **In words:** draw one random sample of 30 fares, compute $\bar{x}$ and s, and use $t_{0.025,\,29}$.
2. **Formula:**
$$\bar{x} \pm t_{0.025,\,29}\,\frac{s}{\sqrt{30}}$$
3. **Example:** the sample gives $\bar{x} = 37.14$ and $s = 49.74$; $t_{0.025,\,29} = 2.045$:
$$\frac{49.74}{\sqrt{30}} = 9.08$$
$$E = 2.045 \times 9.08 = 18.57$$
$$37.14 - 18.57 = 18.57$$
$$37.14 + 18.57 = 55.72$$
(The upper limit is 55.72, not 55.71, because the unrounded margin is used.) The interval is 18.57 to 55.72 pounds. The true mean, 33.30 pounds, lies inside.

The interval is wide because the fares vary enormously ($s = 49.74$) and the sample is small. At 50% confidence, $t_{0.25,\,29} = 0.683$ and the interval shrinks to 30.94 to 43.35 pounds. The 50% range happens to contain 33.30 too, but such an interval misses half the time.

8.1 Several samples: pool them

> **Key point:** Ten samples of 30 are best combined into one sample of 300; averaging their standard deviations and keeping $n = 30$ gives an interval about three times too wide.

Suppose we drew 10 samples of 30 fares. One approach averages the 10 sample means (31.42) and the 10 sample standard deviations (45.88) and builds $31.42 \pm 2.045 \times 45.88/\sqrt{30}$: 14.29 to 48.55 pounds. This interval describes the precision of **one** sample of 30, not of the 300 fares we actually have.

Pooling all 300 fares into one sample and using $n = 300$ gives 26.20 to 36.64 pounds: about a third of the width, and it still contains 33.30. More data should give a narrower interval, and pooling lets it.

Figure 7 puts all four intervals of this case study on one line. Watch the red and green bars: the same 300 fares give a wide interval when averaged the wrong way and a narrow one when pooled.

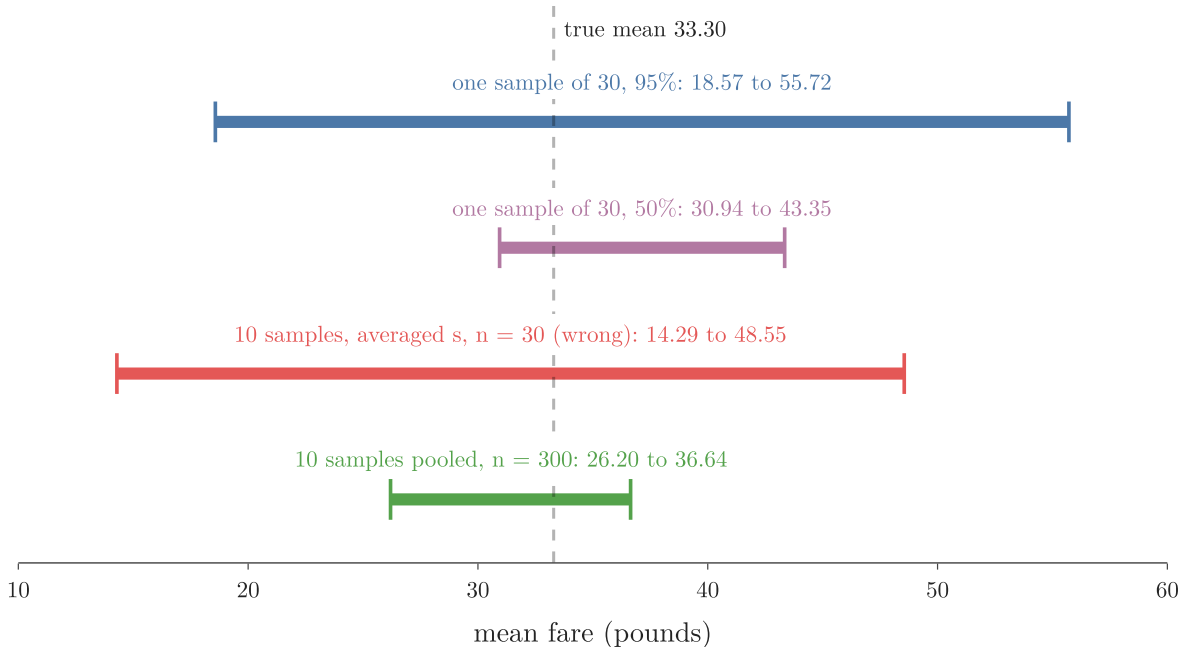


Figure 7: The case study's intervals for the mean Titanic fare: one sample of 30 at 95% and 50%, ten samples of 30 averaged with n = 30 (wrong) and pooled with n = 300; dashed: the true mean 33.30

> **Extra:** The **t-procedure** (G-1938) corrects for the uncertainty of s, not for skewness. The fares are extremely right-skewed (see the Titanic fares). Repeating the one-sample t-interval 4000 times, it contains the true mean only 86.5% of the time for $n = 30$ and 92.8% for $n = 100$. For very skewed data we need larger samples than the usual 30. The bootstrap percentile interval that seaborn uses (see the bootstrap interval) does no better on the same 4000 samples: 86.2% for $n = 30$ and 92.8% for $n = 100$. On data this skewed, the larger sample is what helps.

9. Summary

	Z-procedure	T-procedure
Use when	σ known (rare)	σ unknown (usual)
Spread	σ	s
Distribution	standard normal	Student's t, $df = n - 1$
Interval	$\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$	$\bar{x} \pm t_{\alpha/2,\,n-1}\,s/\sqrt{n}$
95% critical value	1.96	2.262 ($n = 10$), 2.045 ($n = 30$)
Example	25.06 to 30.94 ($n = 100$, $\sigma = 15$)	17.27 to 38.73 ($n = 10$, $s = 15$)

- Use the table's first row to choose: the **t-procedure** (G-1938) is the usual one, because if we do not know μ we almost never know σ, which is computed from the same population.
- Replacing σ by s adds uncertainty; the t-distribution's fat tails account for it, so the interval is wider and honestly claims less precision.
- The t critical value is above $z_{\alpha/2}$ and approaches it as n grows, because s estimates σ better in larger samples.
- "Z with s" gives intervals that are too narrow: 91.9% instead of 95% for $n = 10$, so with s always take the critical value from the t-table.
- For a 95% two-sided interval read the 0.025 column of the t-table, not the 0.05 column, because each tail holds 2.5%; the 0.05 column suits a one-sided interval.
- Pool several samples into one, because averaging their standard deviations and keeping $n = 30$ gives an interval about three times too wide; the t-procedure does not fix strong skewness, so very skewed data needs larger samples (86.5% coverage for the fares at $n = 30$).
- This answers the opening question: when σ is unknown, use s and a critical value from **Student's t-distribution** (G-1906) with $n - 1$ **degrees of freedom** (G-578).

10. Sources

Built from

- CampusX, "Session 44 - Confidence Intervals | DSMP 2023", YouTube, https://www.youtube.com/watc
- Khan Academy, "Introduction to t statistics", YouTube, https://www.youtube.com/watch?v=a2rd4Qy
- Khan Academy, "Simulation showing value of t statistic", YouTube, https://www.youtube.com/watch?v
- Khan Academy, "T-statistic confidence interval", YouTube, https://www.youtube.com/watch?v=hV4p

Other references

- Student [W. S. Gosset] (1908). "The probable error of a mean." *Biometrika* 6(1), 1–25.
- Zabell, S. L. (2008). "On Student's 1908 article 'The probable error of a mean'." *Journal of the American Statistical Association* 103(481), 1–7.
- Casella, G. and Berger, R. L. (2002). *Statistical Inference*, 2nd ed. Duxbury. §5.3.

11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Student's t-distribution (G-1906)	A bell-shaped distribution like the standard normal but with fatter tails. It is used in place of the standard normal when the population's σ is unknown and estimated by the sample's s; it is the distribution of $(\bar{X}-\mu)/(s/\sqrt{n})$ and approaches the standard normal as n grows.
T-procedure (G-1938)	The way to build a confidence interval for a mean when the population standard deviation σ is unknown: use the sample's s and a t critical value in place of z, giving $\bar{x} \pm t_{\alpha/2,\,n-1}\, s/\sqrt{n}$.
Degrees of freedom (G-578)	The number of deviations free to vary, $n-1$ for a sample of size n; it sets the shape of the t-distribution (its parameter).
T critical value (G-1936)	The cut-off from Student's t-distribution that leaves $\alpha/2$ in each tail; the t-interval multiplies the standard error by it in place of $z_{\alpha/2}$. For 95% and $n = 30$ it is 2.045.
T-table (G-1939)	A table of t critical values with one row per degrees of freedom and one column per tail area; we read the cut-off for a t-interval or t-test from it.
Observation (G-1374)	One record of a dataset: one row of the data table.
One-sided and two-sided critical value (G-1383)	The cut-off beyond which the test statistic leads to rejecting H_0 (the critical value): one-sided puts the whole α in one tail, two-sided puts $\alpha/2$ in each of the two tails.